Bow Valley College

# Course Outline

## MATH1901

Math for the Computer Industry

Fall 2020 - Spring 2024

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# MATH1901 Math for the Computer Industry

## COURSE DESCRIPTION

This course provides learners with the practical knowledge and skills in the use of mathematics in relation to computers. Alternate number bases, set theory, logic, and Boolean algebra are the foundation of this course. In addition, learners use basic statistical concepts to understand challenges in information technology.

| | |
|---|---|
| REQUISITES | None |
| EQUIVALENTS | None |
| CREDITS | 3 |
| HOURS | 45 |
| ELIGIBLE FOR PLAR | No |
| ZERO TEXTBOOK COST | |

## COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

**College-Wide Outcomes:**

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

| # | COURSE LEARNING OUTCOME(S) | COLLEGE WIDE OUTCOMES SUPPORTED |
|---|---|---|
| 1 | Apply mathematical concepts to different situations to support problem solving and decision making. | 1, 2, 3, 4, 5, 7 |
| 2 | Apply algebraic concepts to express mathematical formulas and procedures in the computer environment. | 1, 2, 3, 4, 5, 7 |
| 3 | Explain how text and numbers are represented within the computer. | 1, 2, 3, 4, 5, 7, 9 |
| 4 | Use set theory and logic to express decisions carried out by the computer. | 1, 2, 3, 4, 5, 7 |
| 5 | Explain how electronic switches and internal computer logic apply to the construction of digital devices. | 1, 2, 3, 4, 5, 7 |
| 6 | Apply the principles of statistics to collect, describe, and present information technology data. | 1, 2, 3, 4, 5, 6, 7 |

## COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

| WEEK/HOURS | MODULES |
|---|---|
| Week 1 | 1 – Basic Algebra |
| Week 2 | 2 – Number Systems |
| Week 3 | 2 – Number Systems |
| Week 4 | 3 – Computer considerations |
| Week 5 | 3 – Computer considerations |
| Week 6 | 3 – Computer considerations |
| Week 7 | Midterm |
| Week 8 | 4 – Sets and logic |
| Week 9 | Reading week |
| Week 10 | 4 – Sets and logic |
| Week 11 | 5 – Boolean Algebra |
| Week 12 | 6 – Arrays and matrices |
| Week 13 | 6 – Arrays and matrices |
| Week 14 | Final - In Class |

## ASSESSMENT

| COURSE LEARNING OUTCOME(S) | ASSESSMENT | WEIGHT |
|---|---|---|
| 1, 2, 3, 4, 5,6 | Assignments | 30% |
| 1, 2, 3, 4, 5,6 | Group Review Presentation | 5% |
| 1, 2, 3, 4, 5,6 | Formula sheets (to be submitted as part of your midterm and final examinations | 5% |
| 1, 2, 3 | Midterm Exam | 30% |
| 1, 3, 4, 6 | Final Exam | 30% |

Important: For details on each assignment and exam, please see the Course Offering Information.

## PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

## GRADING SCHEME

| **Grade** | **Percentage** | **Grade Point** | **Description** |
|---|---|---|---|
| A+ | 95-100 | 4.0 | Exceptional: superior knowledge of subject matter |
| A | 90-94 | 4.0 | Excellent: outstanding knowledge of subject matter |
| A- | 85-89 | 3.67 | |
| B+ | 80-84 | 3.33 | |
| B | 75-79 | 3.0 | Very Good: knowledge of subject matter generally mastered |
| B- | 70-74 | 2.67 | |
| C+ | 67-69 | 2.33 | |

| | | | |
|---|---|---|---|
| C | 64-66 | 2.0 | Satisfactory/Acceptable: knowledge of subject matter adequately mastered |
| C- | 60-63 | 1.67 | |
| D+ | 57-59 | 1.33 | |
| D | 50-56 | 1.0 | Minimal Pass |
| F | Less than 50 | 0.0 | Fail: an unsatisfactory performance |

## REQUIRED LEARNING RESOURCES

McCullough, R. N. (2006). *Mathematics for computer technology* (3rd ed.). Morton.

SBN: 9780895827005

Additional learning resources may be found in the Course Offering Information or in Brightspace.

## ADDITIONAL INFORMATION

None

Additional information may be found in the Course Offering Information or in Brightspace.

## ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

## INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

**Turnitin:**

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

**Online Exam Proctoring:**

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

**Collection of Personal Information:**

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.